

Lukas Dimitroff

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TECHNICAL PROJECTS

SEND — Hackonomics Hackathon

2026

1st Place, 1,463 participants

- Node-based platform for building and backtesting trading strategies from composable graphs
- Nodes cover data fetches, calculations, and buy/sell conditions; backtests over 6 months of data with daily P/L and replay
- Three-tier architecture: React frontend, Java/Spring REST API, OCaml execution engine over stdin/stdout
- Self-hosted on homelab using Docker, nginx, Cloudflare Tunnel, and PostgreSQL/TimescaleDB

ARM NEON Kernel Optimization

2026 – Present

Contributor (ggml / llama.cpp)

- Hand-writing kernels for IQ4_NL quantization on non-dotprod ARMv8.0-A using Arm NEON SIMD intrinsics
- Identified a missing batched repacked GEMM kernel where ggml falls back to a generic path without SIMD parallelism
- Kernel work in ggml-cpu arm repack.cpp, portable to llama.cpp's ggml dependency

unsloth-turboquant Fork

2026 – Present

Fork maintainer

- Exposed turbo KV-cache quantizations in Unsloth Studio's inference frontend, passing arguments to beellama.cpp
- Built a from-source installer with architecture-aware beellama.cpp compilation

EXPERIENCE

Independent GPU Infrastructure Operator

July 2026 – Present

Vast.ai Host

Irwin, PA

- Operate an RTX 5090 rental host: Ubuntu 24.04 Server, CUDA 13.0, nvidia-container-toolkit
- Isolated renter containers via iptables/nftables egress rules and UniFi VLAN segmentation
- Diagnose driver, networking, and container failures; automated telemetry and recovery maintain uptime

Relight Candles (Nonprofit)

Dec. 2022 – Aug. 2025

Founder

Irwin, PA

- Raised and donated nearly \$10,000 to St. Jude; ran webstore, fulfillment, tax filings, and registration

Norwin's Castle Production Lab (Not-for-profit)

Sep. 2022 – May 2025

Founder / Director of Finance and Operations

Irwin, PA

- Built a webstore and workflows giving special-needs students real work experience; presented financials to administration and School Board

McDonald's

Feb. 2021 – Jan 2023

Crew Member

Irwin, PA

EDUCATION

Northeastern University

May 2028

B.S. Computer Science

Boston, MA

- Entered with 32 credits from dual enrollment and AP; studied abroad in Madrid, first semester

Pennsylvania Governor's School of Science (PGSS)

Summer 2024

Student

Carnegie Mellon University

- Selected 72 of 403 applicants (18%); coursework in CS, discrete math, and AI/ML; researched sensor augmentation for OpenFace occluded-face detection and

Norwin High School

June 2025

National Honors Society; 3.955 unweighted / 4.123 weighted GPA; almost entirely honors and AP core classes

Irwin, PA

SKILLS & ACTIVITIES

Skills: Python; C/C++; OCaml; Java; React; Linux; Docker; Git; CUDA/Triton; networking; UniFi ecosystem

Awards & Activities: Pittsburgh Regional Science & Engineering Fair, Excellence of Presentation; Eurochallenge semifinalist; AFA CyberCamp at Pitt; Westinghouse Honors Science Institute